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Conference Paper · August 2023

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3D Game-Based Approach to Learning Ancient Malay Heritage's History

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Abstract— This paper presents the development of a 3-Dimensional (3D) game highlighting the Ancient Malay culture and history for learning purposes for school children. Many popular digital games feature historical context and aspects. Hikayat Hang Tuah is based on real characters from the Malay Annals. Hang Tuah was a Malaccan royal court warrior during the reign of Sultan Mansur Shah in the 15th century; now embodies the ideal Malay subject and is a symbol of Malay nationalism to this very day. However, the most current generation of students is hardly aware of the history of Hang Tuah and have a lack of interest in learning history because they do not favor learning history from written and lengthy texts. This paper presents a qualitative approach that examines how pupils view history. Such impetuses correspond with the fourth target of the United Nations' Sustainable Development Goals (SDG), which to provide quality education by ensuring inclusive and equitable quality education for all. To increase school kids' interest in studying history, this paper also includes a 3D game that incorporates Malay history. In addition to providing entertainment, it is intended that this 3D game will serve as a platform for learning about the history of the state of Melaka under Sultan Mansur Shah.

Keywords: 3D Games, Game Design, History, Game Based Learning, 3D Modelling, Applications

I. INTRODUCTION

The goal of this study was to strengthen the connections between learning and practise through excellent teaching and learning innovation within a curriculum. Such motivations line up with the fourth Sustainable Development Goal (SDG) of the United Nations, which is to ensure inclusive and equitable quality education for all people (United Nations 2021). In response to the SDG, the Malaysian Ministry of Education (MOE) has been committed to transforming the country's education system in the Malaysian Education Blueprint 2013–2025. In order to maximise their potential, one of the goals outlined in the blueprint is for the students to gain access to high-quality learning that makes use of experiential and technology-

enabled learning models, as well as more individualised and engaging learning experiences. The plan also emphasises the significance of technology-enabled creativity in providing and customizing the educational experience for every student. Moreover, the Malaysian Ministry of Higher Education has mandated the introduction of 'Cybergogy,' virtual-based learning for Learning and Teaching 4.0, in embracing the Fourth Industrial Revolution (MOHE 2018). 3-Dimensional (3D) Game-Based Learning developed in this project is aimed to assist in exploring the history of Hang Tuah in the state of Malacca during the reign of Sultan Mansur Shah in the 15th century. The storyline of this game is adapted from the book "Hikayat Hang Tuah" published in 1908. The legendary tale, Hikayat Hang Tuah, is based on real characters from the Malay Annals. The game proposed in this paper allows the players to experience the 3-Dimensional (3D) environment during the 15th century, highlighting the stories about Malacca's greatest warrior, Hang Tuah, thus essentially learning history through gamification. The game is developed assimilating games like Sekiro: Shadows Die Twice and Assassin's Creed, which are third-person perspective games, where the player shall complete the objective and at the same time explore the environment of the game.

According to the Curriculum Development Division, one of the issues that history teachers frequently encounter is that most students currently have no interest in studying history [1]. The biggest reason is that history books contain many pages, and some individuals who do not favor reading will find it difficult to memorize historical data, which will gradually demotivate them from studying history. In addition, students believe studying history primarily entails listening to lecturers and reading a lot of books, which they find it difficult [1].

Apart from that, there exist traditional elderly and adults who still believe that games will impart bad

influence on younger generations, because there are many games out there with gory and bloody contents that are inappropriate for display, especially to children. This perception is considered by many to affect children's mindset development as they step into adulthood. Research such as [2] shows that some games contain contentious issues like the attraction of games, connections between violence and aggression, as well as causes and effects of games patterns involvement. In contrast to most trends of games, the game presented in this paper introduces the unique and beauty of cultures in Malaysia, mainly the history of Tanah Melayu. The global mindset is majorly accustomed to Western and Northeast Asia history and legends. Teenagers and young people today are increasingly familiar with myths and history from other countries, due to video games like *God of War* and *Age of Mythology*, which bring stories, history, and foreign content into a worldwide setting that appeals to youths [3].

With this study, the author intends to develop a third-person perspective game that can bring gamers, especially teenagers, to discover the origin, life, environment, and history of the Malays. Nowadays, most students dislike reading and even learn about the history of their own country. This is because they often think that learning history is boring, besides requiring a lot of memorizing and reading [1], but have to do so for the sake of examination. By playing this game, they will learn the story of the ancient era in Malacca, as adapted from the book "*Hikayat Hang Tuah*". Students will learn more about the most famous Malay warrior, Hang Tuah; previously, they may have heard only the name, but do not know the story behind it. Other than that, they can anticipate the environment and moment of that era through a 3D perspective. This project also aims to create a game that can raise awareness among young students in Malaysia to appreciate and be proud of the Malay culture, as well as the ancient Malay history in Tanah Melayu and Nusantara. This will trigger further enthusiasm in people to learn about legends and know more about Malay culture as well as culture in Nusantara. Besides, through this game, more people would become aware of Hang Tuah's existence and be proud of the fact that we have our homegrown warrior. People will also appreciate the story of *Hikayat Hang Tuah* which is a particularly national example of the manufactured historical-heroic epic. In history, it is revealed as a biography of a hero in the medieval Malay era [8].

This initiative intends to raise awareness and benefit education through video games. The education in Korea also emphasizes the importance of history education by introducing the concept of game-based learning (GBL), which has been proven its effectiveness through various educational experiments [4]. In Game Based Learning, students are to achieve the learning objective by playing games that blend instructional information and games with little or without supervision by the teacher [6]. According to the researcher, history is the study of the past of an important local unit as it develops into a community, in its context, and in relation to similar local units [7].

Underachievement is one of the more persistent educational problems that kids today deal with, along with learning, behavioural, and emotional challenges. These issues eventually lead to school dropouts [9]. Digital games have potential to be effective instruments for supporting and fostering contextual learning. This detachment of information from a given context facilitates its application to new challenges and topics. Furthermore, the attitude that children or teenagers have toward computer games is the same attitude that we want all our students to have. As a result, it makes sense to try to combine learning content with game motivation. A study finding by the Faculty of Information Science and Technology, Universiti Kebangsaan Malaysia (UKM) shows that games are usually played for the purpose of entertainment. Games are increasingly being blended into school to create a new educational paradigm [2]. Furthermore, games provide an excellent framework for studying cognition in this paradigm of exercises that incorporate computer games. It also can boost for more range of cognitive skills, thinking skills, and creative skills in education, thus computer games essentially offer students a "brain workout" [10].

Computer game-based learning (GBL) is becoming an increasingly popular approach in education. In recent years, there has been a surge of interest in both computer games' potential as educational and instructional tools, as well as research in term of application. The merging of war games, computer science, and operations research, as well as the advent of educational approaches that place a strong emphasis on active, experiential learning and reflection, had led to the birth of educational gaming in the 1950s [11]. Beyond that, most students find it easier to study when they choose their preferred learning style. Students need an interactive learning package because it teaches them how to learn, be independent learners, and communicate with others to obtain relevant knowledge. One of the ways they obtain information and experience is through video games [12]. According to [13], digital games have opened up a whole new way for people to experience or learn history. For instance, a game that can visually immerse players in history is known as a digital war game. In many ways, the medium of digital games is a good instrument for developing historical awareness and understanding. A study by [13] strongly supports the idea of integrating games with teaching as a lifelong game player.

Research by Adam Chapman has proven that game-based learning (GBL) is potential as learning method in the future especially for the next generation [11]. Some scholars have even begun to intensively use historical game studies to cohere into a small but distinct field of interest. More students have considered games to further their study and the source of gaining knowledge in many related topics such as educational science, media studies, narratology, games design, game studies, cultural studies and of course history, heritage, archaeology, and historiography.

Therefore, this research will shed light on the realities and impacts of mobile game-based learning in secondary education on the growth of students' engagement in

learning, particularly their motivation for the topic of history, historical knowledge, and collaborative abilities. [16]. Other than that, simulation, history and computer games nowadays are moving beyond a sole focus on the content of individual historical games in isolation or the application of these games to other domains such as education [17]. According to [18], many people believe digital games are essential future teaching tools to practice and develop high level thinking skills such as problem solving, strategic thinking, resource management, planning and execution, and adapting to changing work settings. As stated by [19], with the significant contribution of software learning, technological evolution has advanced to the point where human ideas are becoming a reality. Supported by another study, game and digital technology proven to be effective in learning [20].

II. DESIGN OF 3D GAME FOR HISTORY OF ANCIENT MALAY HERITAGE

For the development of the game project, ADDIE Model as shown in Fig. 1 had been utilized, since the principal works in a sequential manner, from drawing and designing the levels and characters, to developing Three Dimensional (3D) models and include in the game engine software. To build learning experiences that are efficient, instructional designers and training developers employ the ADDIE Model. The five steps of analyse, design, develop, implement, and evaluate make up a dynamic and adaptive framework for developing effective training and performance support systems.

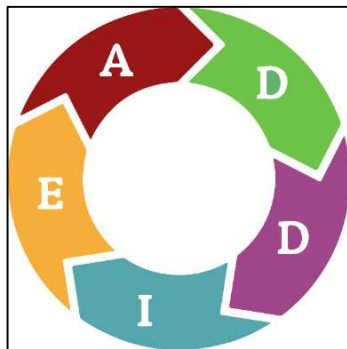


Fig. 1. Addie Model

a) Analysis

The analysis phase consists of gathering the project's issues, and defining the objectives, relevance, talents, and ideas, as simplified in Fig 1, A survey that had been conducted to identify the project's target audience; it is gathered feedback on their opinion of learning ancient Malay heritage through 3D game. In total 183 respondents answered the survey in Google Form. 78% are male respondent and the remaining are female respondent. Respondents are chosen randomly and the invitation to participate in the survey is sent through email. All

respondents are students in secondary school and taking History subjects. 93% of responders are familiar with playing games on their digital device (mobile phone, tablet, computer, or TV). The major findings are presented in Table 1 below:

TABLE I. FINDING FROM SURVEY

Result	Yes	No
Easier to play 3D games and understand the story?	71%	29%
3D games can create interest to learn history?	94%	6%
The 3D game visualize the ancient Malay heritage better compared with images in textbook?	85%	15%
Playing game can relate with the history of Hikayat Hang Tuah?	77%	23%

Based on the major finding, developer brainstormed ideas to visualize the storyline, sketching the characters and level for the game as shown in Fig. 2. The storyboard as shown in Fig. 3 visualizes the flow of the story and the gameplay in the game. The facts from the History subject are included with modification in the game which includes the plot twist, character appearance and environments.

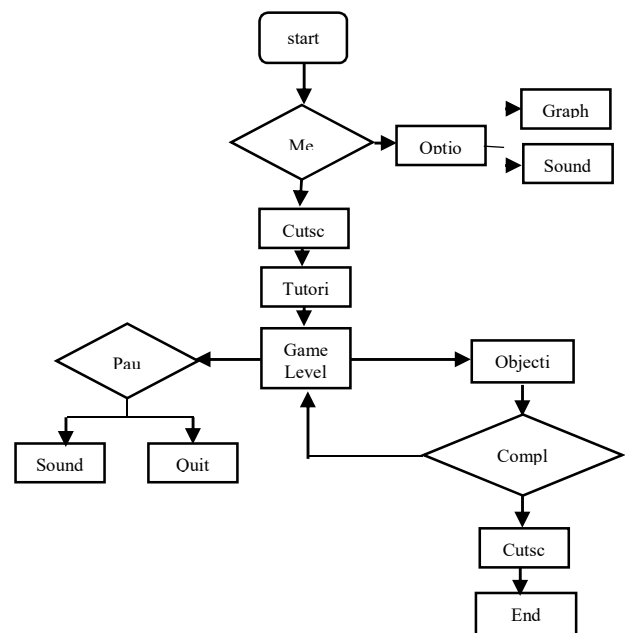


Fig 2. Game Flowchart

Based on the game flowchart the cutscene is the story to inform the user how the history happens. Then the tutorial session is to familiarize the user with the game play. Whenever the user is ready; they can click on Game Level to start playing the game. The challenges are given for the user to obtain their objectives. Once they successfully completed the challenges, a set of cutscene is showing in animation.

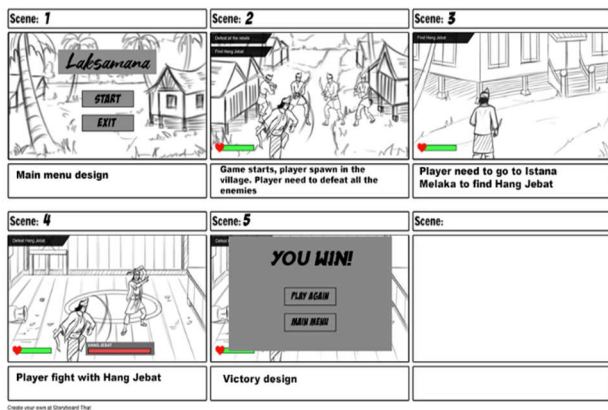


Fig 3. Game Storyboard

b) Design

Learning objectives, content, subject matter analysis, planning, and medium selection are determined during the design process. The storyboard as shown in Fig 3 assisted the visualization of the game and layout. Design framework is then created, inclusive of level design, game mechanics, concept art, and more. The colour scheme, graphic style, and theme used in the game are all influenced by the target audience's preferences, since these factors are vital in defining the game design.

(a)



(b)

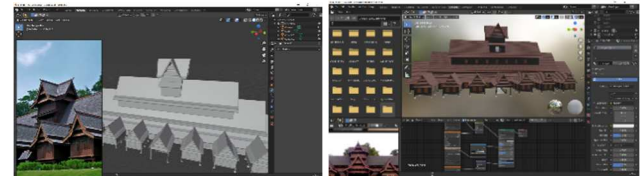
Fig 4. (a) Character Concept Art and (b) 3D Game Environment Sketches

This stage only focuses on creating the concept of the characters and the assets for the game, as shown in Fig 4 (a). Meanwhile, Fig 4 (b) shows the sketches of the character and levels set in Adobe Photoshop. The art style of this game is semi-realism, which is purposely made to accurately reflect the historical background.



(a)

(b)



(c)

(d)



(e)

(f)

Fig 5. 3D Modelling (a-b) exported Character Model in Unreal Engine 4 (UE4), (c-d) process of Istana Melaka in 3D software (Blender) and (e-f) exported from Blender to the Unity software

Fig 5 (a) – (f) show the creation of characters using Zbrush software, using sculpting method and the texturing done in Blender and Substance Painter. After completing, the characters were then exported inside Mixamo and blender to make the rigging and animation. The asset and the level layout for the environment were modelled and textured in Blender according to the concepts.

c) Development

In the development phase, the developer constructs and assembled the content assets that had been developed during the design phase. The three tasks that served to compensate for this period were drafting, production, and assessment. In game production, the content assets developed during the design phase were assembled in the game. Adobe Photoshop, Zbrush, Substance Painter and Blender were used to create and design the game components, as shown in Fig 6, while Unreal Engine was utilized to construct the game. Adobe Photoshop and Adobe Illustrator were used to develop the user interface.

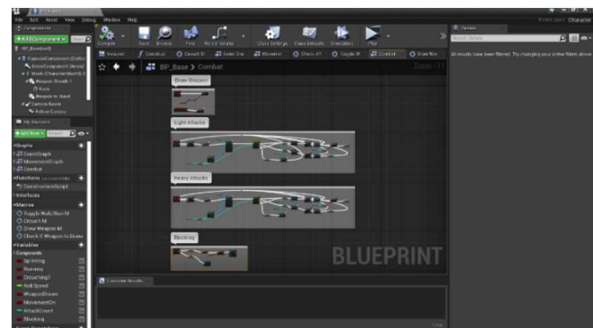


Fig 6: Character Combat Blueprints

The structure draft of the game was made at first, where a default UE4 asset was used for the early gameplay in Unreal. The gameplay and all the interactions such as character movement, inventory, skill and interact with

other object had been programmed using blueprint inside Unreal Engine, as shown in Fig 7.

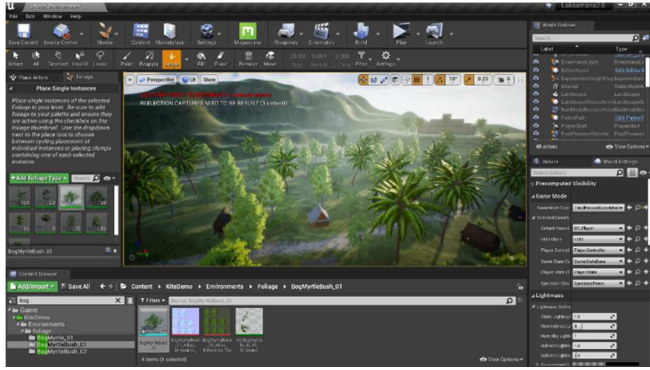


Fig 7: Environment Design in Unreal Engine 4 (UE4)

After completing the 3D modelling process of character, props and making animation, all the assets were transferred and arranged inside Unreal to replace the default asset. Selected sound effects and music were also inserted into the game to make it more like an actual game.

Audio was searched from various websites. One of the websites is opengameart.org. Audio such as sound effects of the weapon swing, footsteps, ambient sound, background music, and others were searched in this stage. The player voice and enemy grunts were taken from Epic store for character vocalization, which were available for free.

d) Implementation

At this stage, the game had been modified and tested often to achieve great efficiency and a favorable game outcome. Internal and external testing was conducted to ensure that the application is easily understood. This phase necessitates continuous redesigning, and enhancement of the game. This phase required to focus on the game environment, characters, and interior to portray the ancient Malay era as shown in Fig 8 (a) – (d) below.



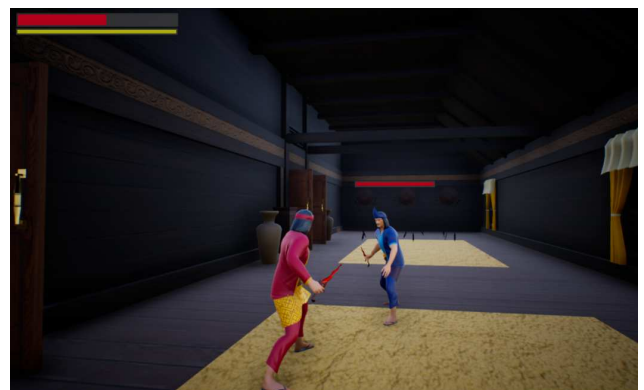
(a)



(b)



(c)



(d)

Figure 8 (a) Malay village (b) Game environment (c) Building in the ancient Malay era and (d) Interior of the building

e) Evaluate

The game tested on school children aged between 10 to 17 years old. The evaluation phase is to ensure that all functions in the game work well. The school children seem to love playing the 3D game, based on their feedback. They answered with assurance that they remembered the game's plot when asked to do so. Each scene in the game can be described in terms of how the landscape reflected the ancient Malay heritage.

III. ANALYSIS

A group of participants are recruited for two sessions. One session to read selected content from textbook and one session for playtesting. The data gathered from the reading

session shows that students read the content in monotone and were less interested in reading the whole content. 3 out of 19 participants completed reading the content but have difficulty answering all questions correctly in verbal. The highest score is 3 out of 10 questions correct. The findings indicated that although they read the article, they did not fully comprehend it.

The data gathered from the playtesting and evaluation session shows the following main findings of deployed the 3D game-based approach in learning the ancient Malay history in children. The playtesting sessions are timed, and the player's behavior is observed. A total of 19 play testers are volunteered to participate. All the participants completed the game, and the average duration is 13 minutes. Furthermore, the highest score obtained is 9 out of 10 questions are correct. Based on the outcome of the playtesting and analysis of the behavior, these are the major findings.

- a. The animation provided in the game relates to the history taught in school or textbook. Thus, they can understand the history better.
- b. The interaction in games enhances the interest in children to keep playing and gaining knowledge.
- c. The challenges in the game gives motivation to them in completing the game.

IV CONCLUSION AND FUTURE WORKS

In a growing number of research, educational games have been shown to be effective in accomplishing educational goals. The conclusion reached from studying the history of Hang Tuah and Melaka in third-person perspective games is that this video game can be used to help educate history. The characters and stories in the video game may not be as nice as anticipated because the resources of the history of the warrior of Melaka are quite scarce and have numerous changes. Making a video game is a lengthy process that should be completed in a group to save time, while achieving the best results. Furthermore, the 3D game-based approach provides a unique opportunity for students to explore and experience the ancient Malay heritage in a virtual environment. This allows for a more hands-on and experiential learning experience, enabling students to visualize and interact with historical artifacts, landmarks, and cultural practices. By immersing themselves in the virtual world, students can develop a deeper understanding and appreciation for the rich history and cultural heritage of the Malay people. Further research can be done with more variety and skill on the characters' attribute, as well as the addition of a plot to complete the story of Hang Tuah. This research can be enhanced further in the future by conducting comparative analysis to assess the effectiveness of 3D game-based learning in comparison to traditional methods of learning historical Ancient Malay heritage. The elements that can assess on the knowledge retention, engagement, and learning outcomes. Other than that, future studies can be enhance on the develop and evaluate appropriate assessment methods and tools for measuring the effectiveness of learning historical Ancient Malay

heritage through 3D game-based learning.

REFERENCES

- [1] Wong Seng Yue, Nor Azan Mat zin, and Azizah Jaafar, "Digital Game-based learning (DGBL) model and development methodology for teaching history," *WSEAS Trans. Comput.*, vol. 8, no. 2, pp. 322–333, 2009.
- [2] M. Z. Nor Azan and S. Y. Wong, "Game based learning model for history courseware: A preliminary analysis," *Proc. - Int. Symp. Inf. Technol. 2008, ITSIM*, vol. 1, 2008.
- [3] M. F. Syahputra, M. K. Widhiyanto, and R. F. Rahmat, "Historical game of majapahit kingdom based on tactical role-playing game," *VISIGRAPP 2017 - Proc. 12th Int. Jt. Conf. Comput. Vision, Imaging Comput. Graph. Theory Appl.*, vol. 1, no. Visigrapp, pp. 327–332, 2017.
- [4] W. H. Lee, S. Y. Park, and W. H. Lee, "Game-Based Learning and Assessment for History Education," no. March, pp. 1–13, 2021.
- [5] J. Hamari, J. Koivisto, and H. Sarsa, "Does gamification work? - A literature review of empirical studies on gamification," *Proc. Annu. Hawaii Int. Conf. Syst. Sci.*, pp. 3025–3034, 2014.
- [6] M. Prensky, "Digital Game-based Learning Prensky," *Games2train*, vol. 1, no. 1, pp. 1–4, 2003.
- [7] I. Of, L. History, and I. N. Secondary, "(the Place and Importance of Local History in Secondary," vol. 6, no. 1, pp. 86–105, 2010.
- [8] Hermansyah Herman, "HIKAYAT HANG TUAH: MALAY EPIC AND MUSLIM MIRROR SOME CONSIDERATIONS ON ITS DATE, MEANING AND STRUCTURE," *Academia*, p. 15, 2018.
- [9] W. Admiraal, J. Huizenga, S. Akkerman, and G. Ten Dam, "The concept of flow in collaborative game-based learning," *Comput. Human Behav.*, vol. 27, no. 3, pp. 1185–1194, 2011.
- [10] L. Davidovitch, A. Parush, and A. Shtub, "Simulation-based learning: The learning-forgetting-relearning process and impact of learning history," *Comput. Educ.*, vol. 50, no. 3, pp. 866–880, 2008.
- [11] N. Whitton, "Motivation and computer game based learning," *ASCILITE 2007 - Australas. Soc. Comput. Learn. Tert. Educ.*, pp. 1063–1067, 2007.
- [12] S. Ma'unah, N. Umamah, S. Sumardi, and R. Afita Surya, "The Enhancement of Attractiveness and Effectiveness of History Learning Using Local History Interactive Teaching Material," *Am. J. Educ. Res.*, vol. 6, no. 11, pp. 1531–1538, 2018.
- [13] K. Schut, "Strategic simulations and our past: The bias of computer games in the presentation of history," *Games Cult.*, vol. 2, no. 3, pp. 213–235, 2007.
- [14] S. Boeykens, "Using 3D design software, BIM and game engines for architectural historical reconstruction," *Des. Together CAADFutures 2011 - Proc. 14th Int. Conf. Comput. Aided Archit. Des.*, pp. 493–509, 2011.
- [15] M. J. Dondlinger, "Educational video game design: A review of the literature," *J. Appl. Educ. Technol.*, vol. 4, no. 1, pp. 21–31, 2007.
- [16] J. F. Hiriart, "crafting gameplay for a digital historical," 2016.
- [17] A. Chapman, A. Foka, and J. Westin, "Introduction: what is historical game studies?," *Rethink. Hist.*, vol. 21, no. 3, pp. 358–371, 2017.
- [18] Y. R. Shi and J. L. Shih, "Game Factors and Game-Based Learning Design Model," *Int. J. Comput. Games Technol.*, vol. 2015, 2015.
- [19] H. A. Safaa N. Saud, M. A. Mohamed Nazar, R. Junaidi Daud, "Arduino Based Quadcopter with Monitoring Camera," *J. Manag. Sci. Univ.*, vol. Vol. 16, no. 1, p. 13, 2018.
- [20] R. Ramli, N. Duriraju and N. Rozzani, "Augmented Reality for Improved User Experience: Himalayan Wildlife Tour Book," *2019 IEEE 9th International Conference on System Engineering and Technology (ICSET)*, 2019, pp. 56–61, doi: 10.1109/ICSEngT.2019.8906388.